

Victoria Nguyen

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SUMMARY

Earlycareer Frontend Software Engineer with fullstack and clinical research background, focused on healthtech solutions. Delivered a React dashboard that reduced data entry time by 20% and built a Python visualization pipeline that improved research reporting accuracy. Deep understanding of healthcare workflows and EMR data enables effective collaboration across multidisciplinary teams. Eager to apply technical and clinical insight to create accessible, efficient software that enhances patient care.

TECHNICAL SKILLS

- **Languages:** Python, SQL (beginner), Bash/CLI, JavaScript, TypeScript, CSS, HTML
- **Frameworks:** Tkinter, Pandas, Matplotlib (basic), React, Node.js, Express, MongoDB, Expo, React Native, JSON, CSV, Streamlit, NumPy
- **Skills:** GUI development, multithreading, file I/O, object-oriented programming, data analysis, version control, SQLite, Rest APIs, SDV, statistical analysis
- **Tools:** VS Code, Jupyter Notebook, Git/GitHub, Notion, EMR

Projects

Synthlab | <https://github.com/vicngan/SynthLab>

Dec 2025 - Present

- Building a privacy-preserving synthetic data platform using Python and Streamlit with three generation methods (CTGAN, GaussianCopula, TVAE) for healthcare research validation.
- Implemented statistical quality assessment including KS testing, correlation analysis, and distribution comparison with interactive Plotly visualizations.
- Designed privacy scoring system with exact-match detection and Distance-to-Closest-Record metrics to quantify re-identification risk.
- Developing PDF literature analyzer with Claude AI summarizer to enable researchers to cross-reference synthetic outputs against published clinical benchmarks.

Heartline Care Companion | https://github.com/vicngan/heartline_companion.py

Oct 2025 - Nov 2025

- Built a mobile care companion for transplant and post-op patients with mood, energy, and symptom check-ins modeled after Michigan Medicine workflows.
- Integrated Twilio SMS alerts that notify caregivers when symptom patterns exceed configurable thresholds.
- Implemented structured vitals logging with JSON/CSV export and Bluetooth LE syncing for clinician review.
- Designed low-friction UX with gentle reminders to reduce alert fatigue and support long-term patient engagement.

WORK EXPERIENCE

Michigan Medicine University Hospital 5C

2025 - Present

Nurse Aide II

Ann Arbor, MI

- Monitored vitals and managed 30+ central lines/drains per shift, providing insight into EMR workflows and clinical data requirements.
- Documented patient data in EPIC using standardized charting, influencing later software designs focused on structured data.
- Worked with interdisciplinary teams in highacuity units to coordinate patient care using the EMR, which improved communication and supported better system integration

Michigan Medicine

2025 - Present

Research Assistant

Ann Arbor, MI

- Analyzed infant feeding behavior videos for the Nipple Coding Study using Observer XT, achieving over 60% reliability, which improved the accuracy of the research data
- Contributed analysis for the ABC Growing Kids project using Python and data visualization, which informed long-term dietary behavior research and facilitated evidence-based development strategy
- Utilized structured datasets and behavioral coding systems, enhancing software data modeling skills, which streamlined data analysis processes, leading to more efficient project completion

EMU Center for Health Disparities

2024 - 2025

Research Assistant

Ann Arbor, MI

- Collected survey data and analyzed it with Excel, uncovering key health disparities that guided community health recommendations; contributed to writing two research summaries and articles that informed local health initiatives; created and translated ten community event flyers using Canva, improving outreach and attendance for CMS Navigator programs; presented a project on Firearm Ownership at the 2025 Spring Symposium, raising awareness among attendees about safety and policy implications

Trinity Health Domino's Farm Urgent Care

2024 - 2025

Medical Assistant

Ann Arbor, MI

- Streamlined workflow and productivity by 30% and decreased averaged patient waiting time by 15%; Supported clinical and lab tasks, enabling accurate data capture for operational and research purposes
- Assisted clinicians with lab work, recording vitals, and procedures, contributing to improved patient care through precise documentation and reliable data handling